

col 703/7203

20th August

Propositional resolution

$\{p_1, \neg p_2, p_3\}$ $\{p_2, p_4\}$

$\{p_1, p_3, p_4\}$

$\{\neg p_3\}$ $\{p_3\}$

$\square$

→ empty clause
(equivalent to false)

What does a derivation look like?

A derivation of a clause C from
a given set of clauses F
is

$C_1, \dots, C_m$
 $\quad \quad \quad \uparrow$

every C_i either comes from F
or is obtained by
resolving C_j and C_k where
 $j, k < i$.

Refutations are the way proofs are done.

$$\{p, \neg q\}, \{q, r\}, \{\neg p, \neg q, r\} \vdash \{r\}$$

$$\{p\}, \{q\} \vdash \frac{p \wedge q}{\neg(p \wedge q) \text{ (KpVnq)}} \neg(p \wedge q)$$

$$\neg(p \wedge q) \vdash \neg p, \neg q$$

$$\neg(p \wedge q) \vdash \neg p, \neg q$$

$$\boxed{p \rightarrow q, q \rightarrow r \vdash p \rightarrow r}$$

C_1 $\square \rightarrow$

C_2

C_3

C_4

$\rightarrow$

$\vdots$

$\square$

either obtained
from premise

or resolving

two
clauses that
are "above"

Soundness

If there is a derivation of $\square$ from a set of clauses F then F is unsatisfiable.

$$F \equiv F \cup \{R\} \quad (\text{from the resolution lemma})$$

$$F \equiv F \cup \{C_1, C_2, \dots, C_m, \square\}$$

$C_1, \dots, C_m$
 $\parallel$
 $\square$

Completeness

If F is unsatisfiable
then there is a derivation of $\perp$
from F .

Notation

Let $F[\text{false}/p]$

be the formula obtained by substituting
false for p in F .

$$F[\text{false}/p_3] = \{ \{p_1\}, \{ \neg p_1, p_2 \}, \{ \neg p_2 \} \}$$

$$F = \{ \{p_1, p_3\}, \{ \neg p_1, p_2 \}, \{ \neg p_2, p_3 \}, \{ \neg p_3 \} \}$$

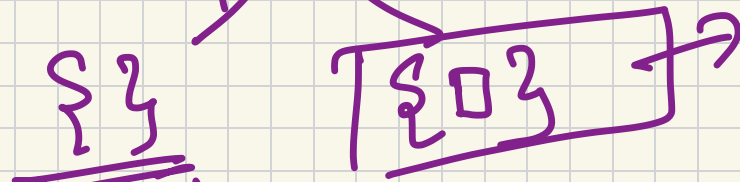
$$F[\text{true}/p_3] = \{ \{ \neg p_1, p_2 \}, \Box \}$$

Proof

by induction on the no. of different propositional variables appearing in F .

Base case

0 propositional vars
two possibilities



one line refutation proof

always true
(can't be the case because F was assumed to be unsat)

Inductive step.

Assume F uses
prop. variables $p_1, p_2, \dots, p_n$

$F[\text{false}/p_n]$

F_0

$F[\text{true}/p_n]$

F_1

Clearly, both F_0 and F_1 are
unsatisfiable.

Therefore there is a derivation of $\perp$
 from F_0 and also from F_1 .
 $F[\text{false}/pn]$

$$\begin{array}{l}
 C_1 \rightarrow F \\
 \boxed{C_1} \rightarrow C_1 \cup \{pn\} \in F \\
 \vdots \\
 \rightarrow \boxed{C_i} \text{ fo } \rightarrow F \\
 \quad \quad \quad \cup C_i
 \end{array}$$

$$\begin{array}{c}
 C_1' \\
 C_2' \\
 \vdots \\
 \perp
 \end{array}$$

$\{p_1\}, \{\neg p_1, p_2\}, \{\neg p_2\}$

F
 $\{\neg p_1, \neg p_2\}$
 $F_0 = F[\text{false}/p_3]$

$\{p_1\} \in F_0$

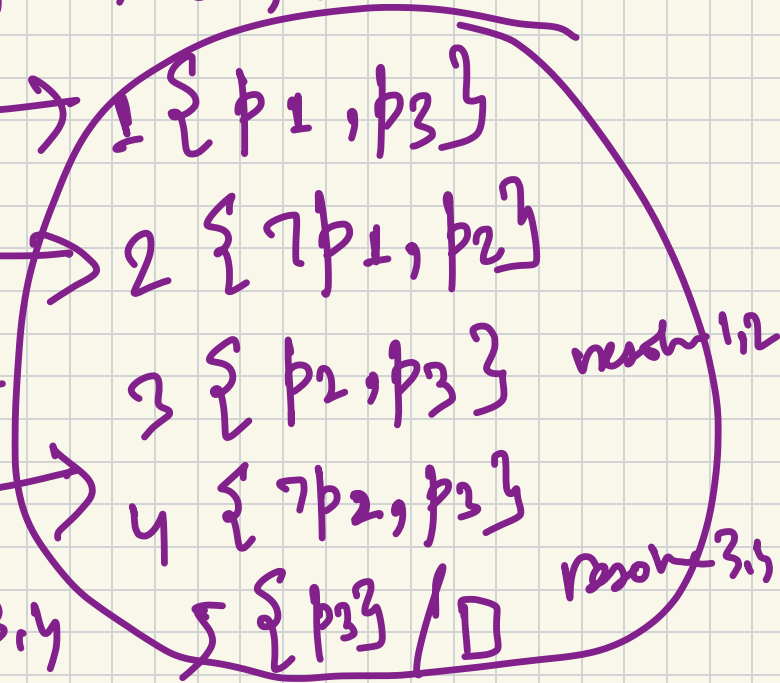
$\{\neg p_1, p_2\} \in F_0$

$\{p_2\}$

$\{\neg p_2\} \in F_0$

$\square$

resolve 3,4



$F = \{ \{p_1, p_3\}, \{\neg p_1, p_2\}, \{\neg p_2, p_3\}, \{\neg p_3\} \}$

F_1

F_2



$C_1 \cup \{p_n\}$

C_1'

C_2'

C_3'

,

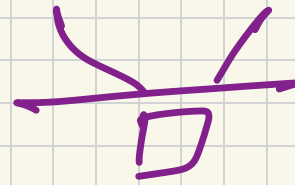
?

$\square$

$p_3/\square$

$\{p_3\}$

$\{p_3\}$



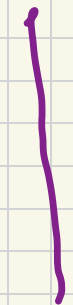
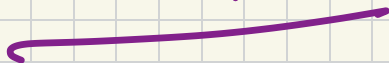
$\{p_n\}$ / $\square$

F

unset

↙ ↘
 $F_0 \equiv F[\text{false}/p]$
unset

$F_1[\text{true}/p]$
unset



Hilbert's proof system

Axioms

$$A1. \quad \phi \rightarrow (\psi \rightarrow \phi)$$

$$A2. \quad (\phi_1 \rightarrow (\phi_2 \rightarrow \phi_3)) \rightarrow \\ ((\phi_1 \rightarrow \phi_2) \rightarrow (\phi_1 \rightarrow \phi_3))$$

$$A3. \quad (\neg \psi \rightarrow \neg \phi) \rightarrow ((\neg \psi \rightarrow \phi) \\ \rightarrow \psi)$$

MP

$$\frac{\phi \quad \phi \rightarrow \psi}{\psi}$$